

Generation Interconnection System Impact Study Report

For Queue Project Q14-0414

Allen 100 kV

May 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0414. The Interconnection Customer proposes a 273.3 MW solar photovoltaic facility interconnecting at the Allen 100 kV substation in the Indiana area. The request was received on December 21, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 224 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,398,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WAYNE TRACE-ALLEN 100kV L	AEP PETTIT AVENUE-ALLEN 100kV	138.2	187	100.0
AEP PETTIT AVENUE-ALLEN 100kV line	WAYNE TRACE-ALLEN 100kV	128.8	201	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WAYNE TRACE-ALLEN 100kV L	Rebuild line with higher-capacity conductor	\$8,156,000	100% (\$8,156,000)
AEP PETTIT AVENUE-ALLEN 100kV	Rebuild line with higher-capacity conductor	\$6,242,000	100% (\$6,242,000)
	Total allocated to project		\$14,398,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.